Peer review of the first research round

Six simulated reviewers • 100 proposals screened • 24 findings
11 September 2026

Recommendation: major revisions to experiment design

The notebook is a useful first brainstorming round. Its central weakness is that several proposed experiments can succeed, fail, or return an expected baseline without testing the mechanism that might advance the open problem. The next revision should sharpen those mechanisms and success criteria before launching more computation.

As an original mathematics submission, it has not established a new contribution. The notebook itself clearly disclaims solved conjectures and verified novelty. Its honesty about scope is a strength; the review does not treat exploratory ideas as failed proof claims.

Priority	Concrete criticism	Required response
1 · Define informative tests	#81 accepts any six of eight inputs; #76 tests only cases where the new derivative terms vanish.	Set an explicit optimization target and include cases that exercise the disputed mechanism.
2 · Check known obstructions	#23 has an impossible trace event; #33 has elementary certificate transfer under a stated hypothesis.	Record the exact obstruction and identify what additional question remains.
3 · Supply real inputs	#8 names no sieve weights; #56 assumes a handle-move sequence absent from its cited proof.	Provide formulas, data and source locations before calling the task executable.
4 · Establish contribution	Related-work comparisons and bridges from small tests to full targets are incomplete.	For selected routes, state the closest result, candidate advance, and missing implication.
5 · Repair reporting	Page 59 merges two run commands; process provenance and pilot status need clarification.	Separate commands, record actual passes, and distinguish completed results from proposed routes.

What held up

Both delivered experiment scripts reproduced their original JSON exactly. Additional implementations matched all union-closed counts, the poset counts through four elements, and the full finite Collatz table. No mathematical error was identified in the completed pilots. Five-element posets were reproduced using the delivered implementation.

This is AI peer review within the same workflow. The four technical reviewers were reassigned across subject sections; the reproducibility reviewer had authored the Collatz audit. These passes provide criticism and crosschecks, with no external expert endorsement.

Scope and evidence

The reviewed submission is the 59-page Math_Frontiers_Round_1.pdf and its companion structured proposals, scripts and outputs. Page references below always refer to that original PDF. The original files were left unchanged.

Reviewer pass	Assignment	Relationship to earlier work
R1 · Editorial	Framing, methodology, status and research readiness	Earlier captain pass; editorial reassessment.
R2 · Number theory	Proposals 1–32, including the Millennium problems	Earlier discrete-mathematics author; cross-section review.
R3 · Discrete mathematics	Proposals 33–55	Earlier geometry/algebra author; cross-section review.
R4 · Geometry and algebra	Proposals 56–80	Earlier computation author; cross-section review.
R5 · Computation and dynamics	Proposals 81–100	Earlier number-theory author; cross-section review.
R6 · Reproducibility	Completed pilots, scripts and run instructions	Earlier Collatz author; same-workflow recheck.

All 100 cards received a conceptual review. Primary sources were checked selectively where they affected a concrete criticism. This is neither a complete literature-status audit nor a formal verification of the paper. The primary assistant assembled the priorities and checked selected arithmetic and source claims.

How to read the findings

Each finding records a location, a short manuscript excerpt, the criticism, an actionable correction and the reviewer’s confidence. A confirmed limitation of a particular test does not rule out its broader research direction. Conditional concerns explicitly identify missing definitions or assumptions. Severity is local to the affected task; it is not an estimate of whether the underlying conjecture is solvable.

Reproduction outcomes

Pilot	Observed outcome	Review coverage
Union-closed families	4,958 eligible families at ground-set size 4; 51 equality cases; minimum maximum frequency 1/2.	Clean rerun and alternative frozenset implementation for sizes 1–4.
Balanced pairs in posets	4,111 non-total labeled orders on 5 elements; each has a pair with probability in [1/3, 2/3].	Clean rerun through size 5; alternative exhaustive implementation through size 4.
Collatz obstruction	The symbolic growing-run identity is correct. All six reported finite counts matched.	Clean rerun plus alternate accelerated-step computation for all 50,000 starts.

R1 · Editorial and methodology

Useful and responsibly labeled brainstorming notebook, with three clearly bounded calibration pilots. Suitable for continued exploratory work after targeted clarification. Not presently a publishable original mathematics research paper: no original contribution is established, and most cards are research prompts rather than specified studies. A methods or expository submission would require its own contribution statement, related-work comparison, and evidence appropriate to that genre. This assessment does not accuse the notebook of claiming proofs or novelty; it explicitly disclaims both.

R1.1 · Originality and related work remain unestablished

Original PDF p. 59 · Major for research publication; acceptable limitation for brainstorming

> *“Claims of new results require a separate literature and proof audit.”*

Criticism. This accurately describes work still missing. General context links plus established/proposed sentences do not identify the closest prior implementation, theorem, or failed route for each proposed mechanism. Consequently neither the pilot results nor the proposed workflow can yet support an original-contribution claim. The notebook should not be submitted as an original mathematics article in this form.

Correction. For each route actually advanced, name the nearest prior result with precise theorem or section references, explain what is reproduced, and state the exact candidate addition. For a methods article, define the methodological contribution and evaluate it against an appropriate baseline; for an expository notebook, retain the current explicit non-novelty framing.

Confidence and limits. High; based on the document itself, not a claim that every proposed idea already exists.

R1.2 · Several experiments omit their central inputs

Original PDF p. 11 · Major before executing the advertised next tasks

> *“For n through one million, evaluate three explicit divisor-based weights, tabulate prime and composite contributions, and document exactly which estimates would be needed to extrapolate.”*

Criticism. The three weights are not given in the card or structured record. Bounding n does not make this an executable protocol: an implementer must still invent the main object being tested. Similar placeholders occur in the precisely specified extension rule on page 32 and five specified parameter choices on page 58. These are legitimate brainstorming prompts, but not yet fully specified experiments.

Correction. Label these tasks protocol design until completed. Before execution, write the actual formulas or parameter lists, input-generation rule, comparison statistic, exactness requirements, and the observation that would reject the mechanism. Do this only for selected routes rather than expanding all 100 prematurely.

Confidence and limits. High; these concrete choices are absent from the reviewed cards and JSON.

R1.3 · The ten-role protocol needs execution provenance

Original PDF p. 2 · Moderate transparency issue

> *“Members first develop approaches independently, then exchange methods and challenge assumptions.”*

Criticism. The notebook establishes ten fictional role descriptions and contains a lead, proposal, and contrary view per item. Those artifacts do not document ten independently generated approaches, the sharing boundary, or an actual sequence of criticism and revision. The sentence can be read as a claimed completed process even though the available record supports a role framework more clearly than compliance with that framework. This is not evidence of ten human participants, and the notebook correctly says all roles are simulated.

Correction. Mark this explicitly as the intended protocol unless execution records substantiate it. For completed rounds, include a concise provenance table of actual AI agent or pass, assigned problems, visible prior material, draft identifier, critic pass, and resulting revision. Distinguish separate generation from statistically independent evidence and avoid implying ten independent reviews per problem.

Confidence and limits. High about absent provenance in the reviewed artifacts; no conclusion about unprovided execution logs.

R1.4 · Activated projects need exact target statements

Original PDF p. 8 · Moderate for standalone research use

> *“003 Birch and Swinnerton-Dyer conjecture”*

Criticism. Problem cards generally give a title followed immediately by an approach, without a formal target statement. For example this card focuses on ranks and numerical L-function behavior without specifying whether the project concerns rank equality alone or the full leading-coefficient formulation. Readers must consult external sources and infer which claim the proposed subproblem is meant to advance. Contrary views help, but do not replace definitions and hypotheses.

Correction. For each activated project add a short exact target with domain, quantifiers, hypotheses, and a separately named restricted subproblem. Identify whether the intended bridge to the full target is a theorem, an assumption, or an unknown step. A brief target line may suffice for the remaining catalog cards.

Confidence and limits. High regarding presentation; no assertion that the proposed restricted work is invalid.

R1.5 · Priorities need budgets and pivot conditions

Original PDF p. 6 · Moderate research-design issue

> *“These are selected follow-up tasks.”*

Criticism. The five priorities have sensible local reasons, but the selection does not compare expected information gain, prerequisite burden, cost, or mathematical impact. The page-three instruction to park routes after repeated uninformative attempts does not specify a round budget or observable stopping condition. Without these, the 100-item ambition can become repeated calibration on convenient examples rather than a focused research portfolio.

Correction. Select one flagship route and one inexpensive calibration route for the next round. For each record an effort or compute cap, exact deliverable, prerequisite dependencies, a falsifier or pivot trigger, and a decision date. Explain the opportunity cost of those choices; keep the other 98 as a catalog, not implied concurrent projects.

Confidence and limits. High; these commitments are absent from the assignment table.

R1.6 · Completed pilots need separate status fields

Original PDF p. 10 · Moderate status-clarity issue

> *“Research proposal; not yet tested”*

Criticism. The Collatz card places this label directly after a Completed pilot paragraph. The distinction is recoverable: the completed obstruction audit differs from the card's proposed residue-class graph. Nevertheless a single status field can be read as applying to the whole card, obscuring which concrete claim has supporting evidence. The same general risk applies when pilot-bearing cards and proposal-only cards share one status schema.

Correction. Split status into proposed-route status and completed-result status. For Collatz, identify the accelerated odd-step obstruction as proved in the notebook with a numerical audit, while explicitly marking the twelve-step residue-graph experiment as unexecuted; link each to its exact statement or output. Preserve the existing non-novelty qualification.

Confidence and limits. High; the nearby labels are visible on page 10.

What this reviewer credited

The opening page plainly calls the work brainstorming and disclaims solutions, verified novelty, universal tractability, invented human participation, and external endorsement.

The three pilots separate a symbolic obstruction from finite exhaustive verification, state their domains, use exact arithmetic, and avoid extrapolating finite success into unrestricted proofs.

Scope limit: I did not independently rerun the pilot scripts for this assignment; another mathematical or computational review is needed to certify their implementation.

R2 · Millennium problems and number theory

Useful, appropriately qualified brainstorming. Three concrete experimental specifications should be clarified before execution. These findings identify one potentially incorrect map interpretation and two unresolved design concerns; they do not refute a claimed solution or the completed Collatz pilot.

R2.1 · Collatz: distinguish two iteration maps

Original PDF p. 10 · #006 Collatz conjecture · moderate

> *“Enumerate residue classes modulo 2^12, derive exact twelve-step affine formulas,”*

Criticism. Unresolved definition ambiguity, with a mathematical error under one natural reading. Page 4 defines the accelerated odd map T(n)=(3n+1)/2^v2(3n+1), but the card does not define its different 'shortcut' map. For the accelerated odd map, a class modulo 4096 does not determine even one affine branch: 1365, 5461, and 9557 are congruent, yet their valuations are 12, 14, and 12 and their outputs are 1, 1, and 7, which are not affine in the inputs. In contrast, the twelve-step claim is valid for the one-division shortcut S(n)=n/2 for even n and (3n+1)/2 for odd n. The completed growing-run argument remains valid.

Correction. State S explicitly in the card and call its iterations shortcut steps, reserving accelerated odd steps for T. Alternatively, keep T but partition adaptively by the full valuation itinerary rather than by a fixed twelve-bit residue class.

Confidence and limits. High about the arithmetic and distinction; the author's intended map is unresolved. The three-value counterexample was checked exactly during this review.

Source: Tao: Almost all orbits of the Collatz map attain almost bounded values

R2.2 · Nonresidues: additive growth needs a contradiction mechanism

Original PDF p. 17 · #021 Vinogradov’s least quadratic nonresidue conjecture · moderate

> *“restricted set can force a contradiction sooner than a direct character-sum estimate”*

Criticism. Unresolved mechanism concern, not a false theorem. Additive growth of a set consisting of quadratic residues does not contradict its multiplicative containment in the residues: sums need not stay quadratic residues or remain smooth. For example, the eight nonzero quadratic residues modulo 17 have sumset equal to the entire field. Thus even maximal expansion is compatible with every original element being a residue. The card's contrary view discusses insufficient expansion and collisions but misses this missing implication. The cited Tao paper supplies distributional and Type II hypotheses, not a conclusion from unrestricted sumset growth alone.

Correction. Specify an additional property that turns an expanded sumset into a contradiction, such as quantitative character cancellation or control keeping relevant sums inside a hypothesized all-residue interval. Define the height bound on smooth integers and include the full quadratic-residue subgroup as a negative control.

Confidence and limits. High that the elementary implication is unavailable and the modulo-17 counterexample is correct; whether a strengthened additive mechanism can work remains open. The sumset was enumerated exactly during this review.

Source: Tao: Vinogradov’s conjecture and Type II estimates

R2.3 · Lang–Trotter: one trace event is impossible

Original PDF p. 18 · #023 Lang–Trotter conjecture · moderate

> *“count traces 0 and 1 at good primes”*

Criticism. Missing local obstruction in the experimental specification, not necessarily a mistaken choice of control. For the second selected curve $E: y^2 = x^3 - 2x + 1$, the cubic factors as $(x-1)(x^2+x-1)$, so $(1,0)$ is a rational point of order two. At every good prime p other than 2 and 5, $E(\mathbb{F}_p)$ has even order and $a_p = p + 1 - \#E(\mathbb{F}_p)$ is even. Consequently the trace-one count for this curve is identically zero, not merely a rare event. This is useful only if explicitly designated as an impossible-event control; otherwise any positive trace-one prediction measures failure to include torsion constraints, not subtle growing-resolution independence.

Correction. Label the second curve's trace-one experiment as a known-zero control and require the mod-two local factor to vanish before collecting data. Use trace two for a non-forced-zero comparison on that curve, after checking its remaining local conditions, or choose a separately verified non-CM curve without this obstruction.

Confidence and limits. High about the exact torsion and parity argument; whether the authors intended a negative control is unstated. Exact point counts at good primes below 101 independently confirmed the predicted parity.

Source: SageMath documentation: Frobenius trace over finite fields

What this reviewer credited

The cards consistently distinguish exploratory diagnostics from proofs and make no novelty claims; finite-range success is not presented as a solution.

The P-versus-NP, Hodge, Yang–Mills, Goldbach, twin-prime, and Lindelöf cards explicitly acknowledge the major gap between their restricted experiments and the unrestricted targets.

Scope limit: The review inspected every card in items 1–32 but selectively checked primary references where they affected a concrete finding; it did not establish the current status of every conjecture again.

R3 · Discrete mathematics

Useful and appropriately qualified brainstorming. Revise one substantive experiment design and clarify three benchmark/input issues before treating the selected next steps as ready to execute. These findings do not allege that the notebook claims solved conjectures or verified novelty.

R3.1 · Hadwiger: ordinary joins already transfer certificates

Original PDF p. 24 · #033 Hadwiger’s conjecture · moderate

> *“Start with seven-chromatic graphs obtained through Hajós joins and search for a short repair rule whenever a join disrupts an existing complete-minor certificate.”*

Criticism. For the ordinary Hajós join of G1 and G2, if the deleted edge of G2 is not a bridge, its endpoints remain connected. That path, followed by the newly added cross-edge, replaces the deleted edge of G1 by an internally disjoint path; contracting it recovers G1 as a minor. Thus a complete-minor certificate in G1 transfers directly. Iterated joins of K7 are therefore a particularly easy certificate benchmark. The paragraph's concern about unpredictable overlap misses that the other input's internal vertices are disjoint. This does not prove Hadwiger for arbitrary graphs: the separate nonadjacent-vertex identification operation in full Hajós constructibility is not covered by this argument.

Correction. First record the elementary certificate-transfer lemma with its non-bridge hypothesis and test the implementation on joins. If the intended research target is the full construction calculus, explicitly add nonadjacent-vertex identification as the difficult operation and distinguish it from a Hajós join. Describe the original finite join experiment as a solved benchmark.

Confidence and limits. High for the stated non-bridge conditional claim: the primary source defines the join on PDF page 2, and the contraction argument is supplied here explicitly. Moderate about the authors' intended generator because its seed graphs and allowed operations are not fully specified.

Source: Primary source: definition of the Hajós join (PDF page 2)

R3.2 · Rota: rank four is a solved benchmark

Original PDF p. 28 · #041 Rota’s basis conjecture · minor

> *“Sample five hundred collections of four bases over the two-element field, identify arrangements stuck under single swaps, and test whether exchanges of length at most four escape.”*

Criticism. The proposed rank-four experiment is a legitimate test of a heuristic, but rank four is already solved for all matroids, not only binary representations. The established/proposed line mentions existing asymptotic results while omitting the exact solved case that contains every proposed instance. Readers need that distinction to interpret successful rearrangements: the experimental question is the geometry of short exchange paths, not whether these instances satisfy Rota's conjecture.

Correction. Label rank four explicitly as a solved benchmark and cite the rank-four computational proof. State the measurable research question as plateau depth or shortest allowed escape length, using a complete solver as ground truth. Do not imply that a successful rearrangement verifies a previously open rank-four case.

Confidence and limits. High: the primary paper states that Rota's basis conjecture holds for all rank-four matroids. The proposal already disclaims a general theorem, so this is a missing contextual distinction rather than a false solution claim.

Source: Kirchweger, Scheucher and Szeider: A SAT Attack on Rota’s Basis Conjecture

R3.3 · Hadwiger–Nelson: identify actual geometric modules

Original PDF p. 28 · #042 Hadwiger–Nelson problem · minor

> *"Choose two published exact-coordinate modules, test twenty explicitly defined placements, verify every asserted distance algebraically, and run bounded five-color SAT searches with independently checkable certificates."*

Criticism. The cited 2025 neural-discovery paper provides valid Hadwiger–Nelson context, but its advertised construction concerns six-colorings of an off-diagonal variant. It is not the direct construction/data citation for the five-chromatic finite unit-distance modules required by this experiment. No two actual modules, coordinate representations, or data locations are identified, so the claimed next action still has an unresolved input-selection step.

Correction. Keep the contextual source if useful, but add a direct finite five-chromatic graph source such as de Grey's paper, identify the two selected graphs and exact-coordinate files, and specify the number field used for distance checking before running placements. The experiment need not change its speculative six-color goal.

Confidence and limits. High about the distinction between the sources: https://arxiv.org/abs/2501.18527 explicitly concerns neural colorings and an off-diagonal variant, while the cited corrective source constructs finite non-four-colorable unit-distance graphs. This is a reproducibility/reference improvement, not a claim that the modules cannot be found.

Source: de Grey: The chromatic number of the plane is at least 5

R3.4 · Unit distances: instantiate inputs and match the statistic

Original PDF p. 29 · #043 Planar unit-distance problem: exact growth · moderate

> *"Extract three reproducible published configurations with at most two hundred points, build exact unit-distance graphs, and compare optimized fifty-vertex subsets against random and symmetry-preserving selections."*

Criticism. The source develops a number-field-tower construction with asymptotic constants; it does not identify three ready-made coordinate configurations of at most 200 points. Thus the next step assumes reproducible small inputs that have not been located or instantiated. In addition, its robust statistic maximizes over the repeated distance separately for each subset. Freezing one unit-distance graph before pruning studies a narrower quantity and can miss the property that motivates using this construction. Neither issue makes pruning invalid, but both affect what this particular test can demonstrate.

Correction. Make construction feasibility the first step: specify one number field, integral basis, box size, and exact point list, or label a small replacement as a toy analogue. For each retained subset, compare both edges at the fixed original distance and maximum multiplicity over all distances, rescaling only after choosing the latter. Treat asymptotic behavior as a later question.

Confidence and limits. High about the source's max-over-distance statistic and asymptotic setup; moderate about availability of small external datasets, which was not exhaustively searched. No claim is made that generating a small coordinate example is impossible.

Source: Source construction and its maximum-over-distance statistic

What this reviewer credited

The section consistently distinguishes proposed searches from universal proofs, including explicit warnings about solver failure, finite sampling, and accidental symmetry assumptions.

Several cards handle target scope carefully: the ordinary Turán tetrahedron problem is separated from its uniform-density variant, and Ryser's conjecture is separated from the disproved Lovász strengthening.

Scope limit: No pilot enumeration was independently rerun, and no proposed search was executed in this review.

R4 · Geometry, topology and algebra

Retain as an explicitly exploratory notebook, with targeted revisions to three cards before treating their next steps as executable pilot assignments. These findings concern source fidelity and whether the proposed tests exercise the intended difficulty; they do not fault the notebook for lacking solutions.

R4.1 · Smooth Poincaré: the cited proof lacks the assumed move sequence

Original PDF p. 36 · #056 Smooth four-dimensional Poincaré conjecture · medium

> *“encode its published handle sequence”*

Criticism. The cited Cha–Kim paper proves standardness using a five-dimensional handlebody comparison, mapping-class arguments, and Gluck twists; its introduction explicitly distinguishes this method from a handle-diagram approach. It does not supply the advertised replayable four-dimensional slide-and-cancellation sequence. Such a sequence might be constructed, but the card treats its availability as a published starting artifact. Moreover, varying a geometric fibered-knot parameter remains inside the already standardized family.

Correction. Either name an actual diagram-level certificate with its theorem and figure references, or change the first task to extracting a structured certificate for the published five-dimensional comparison and its framing/Gluck-twist steps. Treat the geometric family as a validation benchmark, and specify a non-geometric extension separately if an unresolved next case is intended.

Confidence and limits. High for the source-method mismatch and the solved scope: see Theorem A and the introduction's Methods of the proof. This is not a claim that a four-dimensional move certificate cannot exist.

Source: Cha and Kim: Calegari’s homotopy 4-spheres from fibered knots are standard

R4.2 · Chern: the selected tests eliminate the new terms

Original PDF p. 46 · #076 Chern’s isoparametric conjecture for minimal hypersurfaces · medium

> *“evaluate the resulting candidate inequality on known isoparametric examples.”*

Criticism. The proposed obstacle is controlling derivatives of Gauss–Kronecker curvature K after dropping the assumption that K is constant. Isoparametric hypersurfaces in a sphere have constant principal curvatures, so their product K is constant and every newly restored derivative term vanishes. Those examples can check the old identity, but they cannot expose a sign error or inadequate bound in the very terms the proposal aims to control.

Correction. Keep isoparametric examples as zero-derivative regression checks, then add symbolic local-jet tests allowing nonzero derivatives of K while imposing the necessary Gauss–Codazzi and constant mean/scalar-curvature identities. Alternatively, make the first deliverable an explicit estimate for those nonzero terms. Do not require a global non-isoparametric example satisfying the conjecture's hypotheses, which could itself be a counterexample.

Confidence and limits. High. The constant-principal-curvature definition is given in the Cecil–Ryan survey; the vanishing of derivatives of their product is immediate. Local-jet tests would remain only necessary-condition filters.

Source: Cecil and Ryan: survey of isoparametric hypersurfaces

R4.3 · Sphere packing: local D4 improvements are already blocked

Original PDF p. 48 · #080 Four-dimensional sphere-packing problem · medium

> *“test two-point periodic perturbations of a D4 supercell”*

Criticism. If the proposed search means local periodic perturbations seeking a density increase, an established theorem already rules out that outcome: D4 is periodic extreme, including representations with multiple translates of a sublattice. The card mentions local optimality but does not connect it to the proposed experiment, leaving unclear whether this is a negative control or a search expected to improve the packing. Nonlocal, finite-amplitude searches are a different and still legitimate proposal.

Correction. Cite the periodic-extremality result and label the local D4 experiment as a validation test that should find no improvement. For a discovery search, specify a finite-amplitude or multistart periodic family outside the local neighborhood, including which lattice/cell variables change and how all relevant distance constraints will be bounded and certified.

Confidence and limits. High for the mathematical local obstruction; moderate for its application to the intended experiment because the card does not specify perturbation amplitude. Schürmann's introduction and Corollary 11 explicitly cover dimension four.

Source: Schürmann: Perfect, strongly eutactic lattices are periodic extreme

What this reviewer credited

The cards consistently separate established methods, exploratory proposals, and the absence of testing; simulated voices are not presented as real researchers.

Several contrary assessments identify real target-changing errors, including stable versus unstable Andrews–Curtis moves, sliceness versus ribbonness, general cancellation versus affine-space cancellation, and lattice-only versus unrestricted packing bounds.

Scope limit: The review inspected all 25 cards and selectively checked primary sources; it did not audit every cited proof or establish that every proposed method is novel.

R5 · Computation, dynamics and experiment design

Revise the experiment specifications before the next working round. The section is candid about its ambitions and limitations, but some proposed computations have mathematically predetermined outcomes or lack the model definition needed to interpret a result. These are correctable weaknesses in experimental design; they do not establish that the broader research directions are impossible.

R5.1 · P versus BPP: the benchmark has a trivial success condition

Original PDF p. 49 · #081 P versus BPP · Major for the stated experiment

> *“seek one common subset of their eight random strings”*

Criticism. The success condition omits a support-size or seed-length requirement. The entire eight-string set gives zero error immediately. Even interpreting subset as proper does not rescue the benchmark: uniform sampling from any six of the eight strings has total-variation distance 1/4 from uniform sampling from all eight, and therefore acceptance error at most 1/4 for every Boolean function, irrespective of the branching-program restriction. An exact check over all 28 six-element subsets and all 256 Boolean tests confirmed this elementary bound. Thus the proposed search can succeed without detecting any useful pseudorandom structure.

Correction. Define the sampling distribution and optimize support size explicitly, or ask whether a uniform support of at most four strings can achieve the stated error. Specify the branching-program read order, start state, and accepting states. Require an optimum with a lower-bound witness or a certified infeasibility result, then discuss the seed length and composition recurrence. Do not presuppose that a four-string solution exists.

Confidence and limits. Verified design defect. The total-variation calculation and finite exact check are this reviewer's independent verification; the linked book provides pseudorandomness background, not the particular critique.

Source: Wigderson: Mathematics and Computation (background)

R5.2 · Matrix multiplication: direct merging has no candidate groups

Original PDF p. 55 · #093 Matrix multiplication exponent · Moderate

> *“expand its tensor square, group proportional factors”*

Criticism. For the standard exact Strassen seven-term decomposition, the seven factors in each of the three tensor modes are pairwise nonproportional. Its tensor square consequently has 49 pairwise nonproportional factors in each mode: two nonzero pure tensors can be proportional only if the corresponding component factors are proportional. I independently checked the base tensor identity and these projective-factor counts over the rationals. The advertised proportional-factor grouping therefore produces only singleton groups for this natural starting decomposition. Symmetry-related terms are not thereby equal terms. The more general search for identities mentioned in the opening idea might still be productive, but its allowed operations are unspecified.

Correction. Make the standard Strassen tensor square an explicit negative control with the expected result of no direct merges. Specify a bounded search over actual recombinations or re-decompositions of groups of summands, with an exact tensor-identity checker and an objective below 49 terms. Merely applying a common invertible change of basis does not remove the nonproportionality obstruction. Cite a source on the concrete Strassen decomposition and permissible search transformations alongside the general asymptotic-rank reference.

Confidence and limits. Verified for the standard Strassen decomposition and literal proportional-factor merging. This is not a lower bound on the rank of its tensor square and does not exclude new decompositions or more elaborate algebraic identities.

Source: Primary source on the Strassen decomposition

R5.3 · Furstenberg: finite moments cannot exclude a q-grid

Original PDF p. 57 · #098 Furstenberg's ×2, ×3 measure conjecture · Moderate

> *“first without and then with specified Fejér-kernel bounds”*

Criticism. The warning about atomic examples is correct, but the proposed diagnostic has a stronger exact obstruction that should be stated before running an SDP. Let K be the largest Fourier frequency used anywhere in the relaxation, and choose a prime q>K with q coprime to 6. The uniform measure on {j/q: 0≤j<q} is atomic, invariant under doubling and tripling, and has Fourier coefficients identical to Lebesgue measure for every nonzero frequency of magnitude at most K. Consequently every Fejér-kernel integral using only those frequencies, including translated kernels, also agrees exactly. No constraints on that same finite data can distinguish these measures. This atomic measure is not jointly ergodic; the example demonstrates a limitation of the finite relaxation, not a counterexample to Furstenberg's conjecture.

Correction. Add the uniform q-grid as a mandatory analytic negative control, choosing q larger than the actual maximum frequency, including those introduced by moment matrices or invariance relations. If the actual cutoff is 48, q=53 suffices. Treat persistence of this example as predetermined, and focus the experiment on a precisely stated cross-scale inequality. Specify why any added nonconcentration assumption holds for the intended target class and what limit argument could use it. Do not interpret a finite Fejér bound as an atom-exclusion certificate.

Confidence and limits. Verified finite-resolution obstruction by an elementary Fourier calculation. The paper already acknowledges atomic examples; this finding strengthens that caveat into an explicit control and narrows what the experiment can learn.

Source: Source context for the Furstenberg formulation

R5.4 · One-way functions: define the oracle and leakage model

Original PDF p. 51 · #086 Existence of one-way functions · Moderate; model-definition concern

> *“one carefully specified operation revealing information about the encoding”*

Criticism. Whether this operation is an extension depends on which generic-group model is used. In Shoup-style models, algorithms already receive random encoding labels, so revealing a bit of an already visible label adds no information. In Maurer-style models, group elements stay within the oracle and access is mediated by group operations and equality queries. The distinction is documented in Section 1.2 of the linked primary research paper. The proposal does not choose a baseline model or define the leakage, so it is presently impossible to tell whether its comparison changes the information available, merely changes the presentation, or exposes algebraically correlated information. This is a conditional objection: some suitably defined leakage operations would constitute a genuine extension.

Correction. Before enumeration, specify the group order, label distribution, initial information, oracle operations, query budget, strategy grammar, and exact leaked value. Prove that the proposed leakage is not already computable from visible labels, or explicitly label a redundant-leakage test as a calibration. Distinguish exhaustive averaging over secrets from sampling only several encodings, and report the latter as an empirical average rather than a generic-group lower bound. Add a model-specific primary citation.

Confidence and limits. Verified distinction between established generic-group models; verified absence of the required model definition in the card. Redundancy of the eventual leakage is a concern conditional on its still-unspecified definition, not a demonstrated flaw in every possible extension.

Source: Primary comparison of generic-group models, Section 1.2

What this reviewer credited

The section consistently identifies its proposals as untested and makes no novelty claim; its framing is appropriate for a first brainstorming round.

Several contrary views identify the right mathematical obstruction: unrestricted versus restricted circuit models, normalized ground-energy promise gaps versus spectral gaps, and finite renormalization data versus control at every scale.

Scope limit: I read all twenty proposal cards and corresponding paper pages, selectively checked primary sources, and independently performed exact finite checks for items 81 and 93. I did not execute the other proposed experiments.

R6 · Reproducibility and reporting

Finite results reproduced exactly; no mathematical error identified in these pilots. Minor documentation and verification-packaging revisions recommended; one command-formatting defect should be fixed before distribution.

R6.1 · The PDF merges two execution commands

Original PDF p. 59 · moderate

> *“python3 finite_checks.py python3 collatz_audit.py”*

Criticism. Copied as one shell command, this runs only finite_checks.py and passes the remaining tokens as ignored arguments; the Collatz audit does not run. The generator uses a plain newline within a ReportLab paragraph, which collapses it. README commands are correctly separated.

Correction. Render each command in its own preformatted line or paragraph, with explicit line breaks. Verify copying the PDF yields two separate commands.

Confidence and limits. high: extracted PDF text and generator source agree

R6.2 · Exclude the terminal Collatz value

Original PDF p. 4 · minor

> *“every positive odd starting value must already have fallen below its start”*

Criticism. The stated shortcut includes n=1, which is a fixed point and can never descend. The nontrivial obstruction concerns all odd starts n>1. The proof and executable enumeration correctly use starts at least 3.

Correction. Say every odd starting value n>1 must descend below its start within k accelerated odd steps.

Confidence and limits. certain

R6.3 · Add small rejection controls if extending verification

Original PDF p. 59 · minor

> *“The companion archive contains ... the exact experiment scripts and outputs.”*

Criticism. The scripts exhaustively assert the target properties and save aggregate counts and limited witnesses, but do not separately demonstrate that invalid families, nontransitive relations, cycles and excluded cases are rejected. No per-case certificate archive is delivered. This is a robustness gap, not evidence of incorrect counts; alternative implementations reproduced the tested counts.

Correction. Add a small suite of explicit accepted/rejected fixtures, and save a verifier or compact per-case manifest if later claiming machine-checkable certification. Do not describe the current output as a formal certificate.

Confidence and limits. high

R6.4 · Record the execution environment and source hashes

Original PDF p. 59 · minor

> *“The experiments require only Python's standard library.”*

Criticism. This dependency claim is correct, but the report lacks the exact Python version and source hashes. Reproduction succeeded under Python 3.12.14. Random seeds are unnecessary because completed pilots are exhaustive and deterministic; seeds for unexecuted proposed experiments are outside this pilot claim.

Correction. Add Python version, platform, source SHA-256 hashes and an explicit statement that no random generator is used. Keep claimed experimental reproducibility separate from rebuilding the PDF, whose renderer uses additional packages and a runtime-specific font path.

Confidence and limits. high

What this reviewer credited

Both delivered scripts ran successfully from clean temporary copies and reproduced their JSON objects exactly.

All displayed pilot counts agree with supplied outputs.

Scope limit: Alternative poset implementation covered n=2,3,4; n=5 was reproduced through the supplied script and reviewed algorithmically, not independently exhaustively reimplemented.

Revision plan and review record

The next revision should turn a small selection of these ideas into interpretable studies. Adding more simulated enthusiasm or more numerical examples will not resolve the gaps identified here.

1. Correct the concrete defects

Fix the run commands and n>1 wording. Define the two Collatz maps. Label the elliptic-curve trace-one event, the Strassen direct-merge test and local D4 perturbations as controls with known expected outcomes.

2. Write one-page executable protocols

For each chosen study, provide the exact target and restricted claim, formulas or input files, baseline, resource cap, statistic, expected control behavior and a result that would force a change of direction. Resolve the missing weights, parameters and oracle definitions before running them.

3. Test the proposed mechanism

For #81, constrain or minimize sample size. For #76, analyze admissible nonzero derivative terms. For #21, supply the implication from additive behavior to a contradiction. For #98, use the q-grid obstruction to identify what a cross-scale statement must add.

4. Position the work against the literature

For activated projects, cite the closest theorem and algorithm precisely. Distinguish reproduced baselines from candidate additions, and say which step could connect a finite study to the full problem. Only a separate proof and novelty audit could support an original-result claim.

5. Commit to a bounded next round

Choose one principal project and one inexpensive control study. Record effort limits, dependencies, pivot criteria and decision points. The 100-card catalog remains useful for future selection; it does not itself justify 100 simultaneous projects.

Artifact and execution record

Reviewed PDF SHA-256: b7749f1a1d74ecdeb7a59d8d5dce0d39 8f6b710b8374bea8e9a7ff89eaf477b7

Reproduction environment: Python 3.12.14; Linux-6.18.35-x86_64-with-glibc2.39. Completed pilots are deterministic and use the standard library.

finite_checks.py SHA-256: 0e892da5473acfcf4096e4915cef17c8 61afd765e6f7dbda0e597ecc2d4a2a21

collatz_audit.py SHA-256: f635aff55b07e7b0a12ca8499f61f8f4 e8fa7d76bb4f115362992bd67f40535b

The verdict and correction priorities are editorial judgments assembled from the six AI reports. Exact local calculations support several particular objections; shared model assumptions and incomplete literature coverage remain possible sources of error. The review establishes no solution of an open problem.